PHYS 365 — Formula Sheet Guide

Every formula on the posted course formula sheet (*PHYS 365 - Formula Sheet.pdf*), plus the relativistic Doppler formula from the midterm sheet. For each one: what every symbol means, when the formula is valid, what kind of question needs it, and which past exam questions used it.

How to read the question codes: M = midterm, F = final, followed by the year. So **M25 Q9** is Midterm Fall 2025, Question 9, and **F23 Q22** is Final Fall 2023, Question 22. The worked solutions for every one of these are in the *Midterm Practice Solutions* and *Final Exam Practice Solutions* folders.
Midterm sheet = everything up to and including the optics section, plus relativistic Doppler. It has no modern physics.
Anything in **bold** that says **not on the sheet** is something you have to memorize.

Oscillations (simple harmonic motion)

These describe anything that moves back and forth with a restoring force proportional to displacement: a mass on a spring, a pendulum, or one point on a wave.

SHM position

$$x(t) = x_m \cos(\omega t + \phi)$$

Variables	x = displacement from equilibrium (m). x_m = amplitude, the largest displacement (m). ω = angular frequency (rad·s^{-1}). t = time (s). ϕ = phase constant (rad), which sets where the motion starts. $\omega t + \phi$ is the phase.
Assumptions	Ideal SHM: the restoring force is $F = -kx$, with no friction or damping, so the amplitude stays constant.
Use it for	Reading or writing $x(t)$ from a graph, finding ϕ from the starting conditions, and finding position at a given time. Differentiate it yourself to get velocity $v = -\omega x_m \sin(\omega t + \phi)$ and acceleration $a = -\omega^2 x$. **Those two are NOT on the sheet.** To find ϕ, use $x(0) = x_m \cos\phi$ and pick the root of $\cos^{-1}$ whose velocity sign matches the graph.
Past questions	M25 Q4, Q13 · M24 Q10 · M23 Q8 · M22 Q1, Q2 · F25 Q6 · F24 Q10 · F23 Q1, Q2 · F22 Q5

Angular frequency, frequency, and period

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Variables	f = frequency, cycles per second (Hz). T = period, time for one cycle (s). ω = angular frequency (rad·s^{-1}).
Assumptions	Always true for any periodic motion or wave.
Use it for	Converting between ω, f, and T. Watch the trap where an answer gives ω when the question asks for f (M24 Q2). Also used to read f from a wave equation (F25 Q10).
Past questions	Nearly every oscillation or wave question: M25 Q4 · M24 Q2, Q11 · F25 Q10 · F24 Q13 · F23 Q18

Mass on a spring

$$\omega = \sqrt{\frac{k}{m}}$$

Variables	k = spring constant (N·m^{-1}). m = oscillating mass (kg).
Assumptions	Ideal massless spring obeying Hooke's law, no friction. ω does not depend on amplitude. With several springs, use the effective k: springs on both sides of the mass (parallel) give $k_1 + k_2$. Springs joined end to end (series) give $\frac{k_1 k_2}{k_1 + k_2}$. **Neither combination rule is on the sheet.**
Use it for	Finding the frequency or period of a block on springs, or showing that ω is the same in every trial for the same system.
Past questions	M25 Q10 · M24 Q11 (parallel) · M23 Q3 · F23 Q18 (series)

Simple pendulum

$$\omega = \sqrt{\frac{g}{\ell}}, \qquad T = 2\pi\sqrt{\frac{\ell}{g}}$$

Variables	g = gravitational acceleration (9.81 m·s^{-2} on Earth). ℓ = length from pivot to the centre of the bob (m).
Assumptions	Small angles ($\sin\theta \approx \theta$), a point-mass bob, a massless string, and no air resistance. **The mass does not appear**, so the period does not depend on it.
Use it for	Pendulum on another planet or the Moon ($T \propto \frac{1}{\sqrt{g}}$), changing the length, and conceptual questions about mass.
Past questions	M24 Q9 · F25 Q4 · F24 Q11

Waves: describing a travelling wave

Wave equation

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$$

Variables	$y(x,t)$ = displacement of the medium. v = wave speed.
Assumptions	Linear, non-dispersive medium (all frequencies travel at the same speed). Any function $f(x \pm vt)$ solves it.
Use it for	Mostly conceptual: checking whether a function can be a wave, or reading off v by comparing the two sides.
Past questions	Not directly tested in past exams. Know what it means.

General travelling wave

$$y(x,t) = f(x \pm vt)$$

Variables	f = any shape (a pulse). $+vt$ means the wave moves in the $-x$ direction. $-vt$ means it moves in the $+x$ direction.
Assumptions	The shape does not change as it travels (no dispersion or damping).
Use it for	Pulse problems: shift each pulse by $v\,\Delta t$, then add the displacements (superposition). Also identifying which function is a wave moving in a given direction at a given speed: factor it as $f(x + vt)$.
Past questions	M23 Q5 · M22 Q5 · F24 Q9 · F22 Q7

Sinusoidal travelling wave

$$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$$

Variables	y_m = amplitude. k = wavenumber (rad·m^{-1}). ω = angular frequency. ϕ = phase constant. The sign rule is the same as above ($-\omega t$ moves in $+x$). Older sheets write it with $\sin$. The two differ only by a phase of $\frac{\pi}{2}$.
Assumptions	Harmonic wave in a linear medium.
Use it for	Reading k, ω, λ, f, and v off an equation, matching equations to graphs, and writing an equation from a $y(t)$ graph at $x = 0$. Use $\sin\left(\theta \pm \frac{\pi}{2}\right) = \pm\cos\theta$ from the math section to switch between sin and cos.
Past questions	M23 Q2, Q12 · M22 Q3 · F25 Q10 · F23 Q16 · F22 Q2, Q6

Wave speed

$$v = \frac{\omega}{k} = f\lambda$$

Variables	v = wave (phase) speed. λ = wavelength.
Assumptions	Always true for a sinusoidal wave. **Key idea:** v is set by the **medium** and f by the **source**, so changing f changes λ, not v.
Use it for	Comparing wave speeds from equations, and conceptual questions about what changes when frequency or tension changes. At a boundary f stays the same, so λ and k change.
Past questions	M25 Q1, Q21, Q22 · F25 Q1, Q2 · F23 Q3, Q4 · F22 Q1

Wavenumber

$$k = \frac{2\pi}{\lambda}$$

Variables	k = wavenumber (rad·m^{-1}). λ = wavelength (m).
Assumptions	Always true.
Use it for	Reading λ from an equation or a graph (a larger k means a shorter wavelength).
Past questions	M23 Q2, Q12 · F25 Q10 · F22 Q2

Light in a medium

$$v = \frac{c}{n}, \qquad \lambda = \frac{\lambda_0}{n}$$

Variables	c = speed of light in vacuum. n = index of refraction ($n \geq 1$). λ_0 = vacuum (air) wavelength. λ = wavelength inside the material.
Assumptions	Frequency stays the same when light enters a material. Only the speed and wavelength shrink.
Use it for	Thin films (the wavelength inside the film is $\frac{\lambda_0}{n}$), and counting how many wavelengths fit in a layer: $N = \frac{Ln}{\lambda_0}$ (not on the sheet).
Past questions	M23 Q13, Q16 · M22 Q11, Q12 · F24 Q5 · F22 Q13, Q14

Waves: speed in different media

Wave on a string

$$v = \sqrt{\frac{\tau}{\mu}}$$

Variables	τ = tension (N). μ = linear mass density $= \frac{m}{L}$ (kg·m^{-1}). $\mu = \frac{m}{L}$ **is not on the sheet.**
Assumptions	Small transverse displacements and a uniform, perfectly flexible string. A hanging mass gives $\tau = Mg$ (not on sheet). In a hanging rope, the tension at a point equals the weight below it.
Use it for	Guitar and standing-wave problems (combine with $f_n = \frac{nv}{2L}$), finding the tension needed for a note, and comparing strings of different thickness ($\mu \propto D^2$ for the same material).
Past questions	M25 Q12 · M24 Q2, Q4, Q12, Q20 · M23 Q4 · M22 Q14 · F25 Q8, Q12 · F23 Q6, Q12 · F22 Q1, Q3, Q6

Sound in a gas

$$v = \sqrt{\frac{\gamma R T}{M}}$$

Variables	γ = ratio of specific heats: $\frac{5}{3}$ monatomic (He, Ar), $\frac{7}{5}$ diatomic (N_2, O_2, air). R = 8.314 J·mol^{-1}·K^{-1}. T = temperature in **kelvin**. M = molar mass in **kg·mol^{-1}** (air is 0.0288).
Assumptions	Ideal gas with adiabatic compressions. Convert °C to K and g to kg. For a mixture, $M = \sum(\text{fraction})M_i$ (not on sheet).
Use it for	Speed of sound vs. temperature or gas type, finding the air temperature from a measured speed, and ratios between gases (R and T cancel).
Past questions	M25 Q8 · F24 Q14

Longitudinal waves in a solid rod

$$v = \sqrt{\frac{Y}{\rho}}$$

Variables	Y = Young's modulus, the stiffness under stretching (Pa). ρ = density (kg·m^{-3}).
Assumptions	Thin rod, small strains (linear elastic).
Use it for	Speed of sound along a rod or wire made of a given material.
Past questions	Not used in past exams.

Sound in a fluid

$$v = \sqrt{\frac{B}{\rho}}$$

Variables	B = bulk modulus, the resistance to compression (Pa). ρ = density.
Assumptions	Small compressions in a liquid or gas.
Use it for	Combine with the bulk modulus definition (below) when a piston compresses a liquid. Also used for impedance $Z = \sqrt{\rho B}$.
Past questions	M24 Q6

Waves: impedance, energy, intensity, and sound level

String impedance

$$Z = \sqrt{\mu\tau} = \frac{\tau}{v} = \mu v$$

Variables	Z = mechanical impedance of the string (kg·s^{-1}), meaning how hard it is to move.
Assumptions	Joined strings under the **same tension** (they are pulled by the same force). Then $Z \propto \sqrt{\mu}$.
Use it for	Any joined-string reflection or transmission problem: compute $\frac{Z_2}{Z_1}$ first, then use R, T, R_e, and T_e.
Past questions	M25 Q3, Q11 · M23 Q17 · F25 Q3 · F23 Q6, Q14

Acoustic impedance

$$Z = \sqrt{\rho B} = \rho v$$

Variables	ρ = density. B = bulk modulus. v = sound speed in that medium. Units: Pa·s·m^{-1} (rayl).
Assumptions	Plane sound waves in a fluid.
Use it for	Sound crossing a boundary (water to liquid, air to foam), designing non-reflecting materials ($Z_1 = Z_2$), and how Z changes with altitude.
Past questions	M24 Q15, Q16 · M22 Q13 · F25 Q13 · F24 Q2, Q15 · F22 Q15

Average power on a string

$$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$$

Variables	P_{avg} = average power carried along the string (W). y_m = amplitude.
Assumptions	Sinusoidal wave, averaged over a period. Note that $P \propto y_m^2$ and $P \propto \omega^2$.
Use it for	Finding the power, or the frequency needed for a given power. Ratio questions such as "amplitude up by 22.5%, so energy goes up by $1.225^2 = 1.5$".
Past questions	M25 Q23 · M24 Q2 · F25 Q12

Pressure amplitude vs. displacement amplitude

$$\Delta p_m = (v\rho\omega)s_m$$

Variables	Δp_m = pressure amplitude (Pa), the largest change from normal pressure. s_m = displacement amplitude of the air molecules (m). $v\rho = Z$.
Assumptions	Sinusoidal sound wave. Δp_m is **directly** proportional to s_m.
Use it for	Converting between how much the air moves and how much the pressure changes (such as finding s_m for a 20 mPa sound).
Past questions	M25 Q20 · M24 Q5 · F24 Q1

Pressure wave

$$\Delta p = \Delta p_m \sin(kx - \omega t)$$

Variables	Δp = gauge pressure change at x, t. A negative value is a rarefaction (expansion).
Assumptions	It goes with displacement $s = s_m \cos(kx - \omega t)$, which is **not on the current sheet**. Pressure is 90° out of phase with displacement: maximum displacement means zero pressure change.
Use it for	Conceptual true/false questions about pressure and displacement.
Past questions	M24 Q5

Sound intensity

$$I = \frac{1}{2}\rho v\omega^2 s_m^2$$

Variables	I = intensity, power per area (W·m^{-2}).
Assumptions	Sinusoidal sound wave. Combined with $\Delta p_m = v\rho\omega s_m$ it gives $I = \frac{(\Delta p_m)^2}{2\rho v}$ (not on sheet).
Use it for	Going between dB level, intensity, and pressure or displacement amplitude.
Past questions	F24 Q1

Intensity of a point source

$$I = \frac{P}{4\pi R^2}$$

Variables	P = source power (W). R = distance from the source (m). $4\pi R^2$ = area of a sphere.
Assumptions	Isotropic point source (it radiates equally in all directions) and no absorption. So $I \propto \frac{1}{R^2}$.
Use it for	How intensity or dB changes with distance. Ratio form: $\frac{I_1}{I_2} = \frac{R_2^2}{R_1^2}$.
Past questions	M22 Q15

Intensity, general

$$I = \frac{P}{A}$$

Variables	A = area the power spreads over. For a line source it is a cylinder, $A = 2\pi r L$ (not on sheet).
Assumptions	Power spread uniformly over the area.
Use it for	Line sources, beams, and solar panels ($P = IA$).
Past questions	F23 Q7, Q19

Sound level

$$\beta = (10\ \text{dB}) \log_{10}\left(\frac{I}{I_0}\right)$$

Variables	β = sound level (dB). $I_0 = 1 \times 10^{-12}$ W·m^{-2} is the hearing threshold (0 dB).
Assumptions	Logarithmic scale. Every $imes10$ in intensity adds 10 dB.
Use it for	Converting dB to intensity: $I = I_0 10^{\frac{\beta}{10}}$.
Past questions	M22 Q15 · F24 Q1

Change in sound level

$$\Delta\beta = (10\ \text{dB}) \log_{10}\left(\frac{I_2}{I_1}\right)$$

Variables	Same variables as above.
Assumptions	I_0 cancels, so you only need the ratio.
Use it for	"Intensity drops by a factor of 100, so the level drops by 20 dB." Combine with the spreading law below.
Past questions	M23 Q18 · M24 Q13

Spreading plus absorption

$$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)\, e^{-\alpha \Delta r}$$

Variables	n = number of dimensions the wave spreads in: 3 for a point source (sphere), 2 for a line source (cylinder), 1 for a plane wave or tube. α = attenuation (absorption) coefficient (m^{-1}). $\Delta r = r_2 - r_1$.
Assumptions	Uniform medium. For a point source in air use $n = 3$. **Recognizing which n to use is not stated on the sheet.**
Use it for	Sound level at a far distance including air absorption. Take $10 \log_{10}$ of the ratio, using $\log_{10}(e^x) = 0.4343x$.
Past questions	M24 Q13

Waves: reflection and transmission at a boundary

Amplitude reflection and transmission

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \qquad T = \frac{2Z_1}{Z_1 + Z_2}$$

Variables	Z_1 = impedance of the medium the wave comes **from**. Z_2 = the medium it goes **into**. R = reflected amplitude ÷ incident amplitude. T = transmitted amplitude ÷ incident amplitude. (Older sheets use lower-case r and t.)
Assumptions	Normal incidence. $R < 0$ means the reflected wave is inverted (going into a heavier or stiffer medium, $Z_2 > Z_1$). T is always positive and can be larger than 1.
Use it for	Joined strings, the amplitude of a transmitted wave, beam splitters, and insulation design ($R \approx 0$ absorbs, $\lvert R \rvert \approx 1$ blocks).
Past questions	M25 Q5, Q11 · M24 Q15, Q16 · M22 Q13 · F25 Q13 · F23 Q6

Energy (power) reflection and transmission

$$R_e = R^2, \qquad T_e = \frac{Z_2}{Z_1} T^2, \qquad R_e + T_e = 1$$

Variables	R_e = fraction of power reflected. T_e = fraction of power transmitted.
Assumptions	Energy is conserved at the boundary. Do not just square T: you must multiply by $\frac{Z_2}{Z_1}$. With $r = \frac{Z_2}{Z_1}$, $T_e = \frac{4r}{(1+r)^2}$.
Use it for	Any question that says "power" or "energy" transmitted, and finding Z_2 from a given percentage (this can lead to a quadratic).
Past questions	M25 Q3 · M23 Q17 · F24 Q2 · F23 Q14 · F22 Q15

Bulk modulus

$$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$$

Variables	Δp = pressure increase. $\frac{\Delta V}{V}$ = fractional volume change (negative under compression, so $B > 0$).
Assumptions	Small, elastic compression. For a cylinder of fixed radius, $\frac{\Delta V}{V} = \frac{\Delta L}{L}$. With fixed mass, $\frac{\Delta V}{V} = -\frac{\Delta \rho}{\rho}$.
Use it for	Piston compressing a liquid, which is then used in $v = \sqrt{\frac{B}{\rho}}$.
Past questions	M24 Q6 · F24 Q15

Young's modulus

$$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$$

Variables	$\frac{F}{A}$ = stress (force per cross-sectional area). $\frac{\Delta L}{L}$ = strain (fractional change in length).
Assumptions	Linear elastic region.
Use it for	Finding Y for $v = \sqrt{\frac{Y}{\rho}}$.
Past questions	Not used in past exams.

Pressure from displacement

$$\Delta p = -B\frac{\partial s}{\partial x}$$

Variables	$s(x, t)$ = air displacement.
Assumptions	Small-amplitude sound.
Use it for	This is where $\Delta p_m = v\rho\omega s_m$ comes from. It explains why pressure is 90° out of phase with displacement.
Past questions	Conceptual only.

Wave interference, standing waves, and Doppler

Two equal travelling waves added

$$y'(x,t) = \left[2y_m \cos\left(\tfrac{1}{2}\phi\right)\right] \sin\left(kx - \omega t + \tfrac{1}{2}\phi\right)$$

Variables	ϕ = phase difference between the two waves. The bracket is the resultant amplitude.
Assumptions	**Only for equal amplitudes**, same k and ω, and the same direction. For unequal amplitudes use phasors instead.
Use it for	Finding the resultant amplitude, or solving $2y_m \cos\left(\frac{\phi}{2}\right) = \text{target}$ for ϕ. Remember to double $\frac{\phi}{2}$ at the end.
Past questions	M25 Q17 · M24 Q3, Q8 · F24 Q12

Standing wave

$$y'(x,t) = [2y_m \sin(kx)] \cos(\omega t)$$

Variables	Two equal waves travelling in **opposite** directions. Nodes are where $\sin(kx) = 0$ ($x = 0, \frac{\lambda}{2}, \lambda, \ldots$). Antinodes are where $\lvert\sin(kx)\rvert = 1$ ($x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \ldots$).
Assumptions	This form has a node at $x = 0$.
Use it for	Choosing the right standing-wave equation from a graph, and finding node or antinode positions.
Past questions	F25 Q9

Phase difference from path difference

$$\Delta\phi = \frac{\Delta L}{\lambda} 2\pi$$

Variables	ΔL = difference in distance travelled. $\Delta\phi$ = resulting phase difference (rad).
Assumptions	Same wavelength along both paths.
Use it for	Two speakers or rays, finding λ (and from it v and T) from a measured phase difference, and noise cancelling.
Past questions	M23 Q13 · F24 Q13, Q14

Beats

$$f_{beat} = f_1 - f_2$$

Variables	f_1, f_2 = two close frequencies. Use the absolute value.
Assumptions	Two nearly equal frequencies heard together. You hear the average tone $\dfrac{f_1 + f_2}{2}$ (not on sheet), getting louder and softer at f_{beat}.
Use it for	Tuning pipes or strings, the frequency of a second speaker (keep v fixed, so k must change), and Doppler-shifted beats.
Past questions	M25 Q22 · M24 Q17 · F25 Q5

Doppler effect (sound)

$$f' = f\,\frac{v \pm v_D}{v \pm v_S}$$

Variables	f = emitted frequency. f' = detected frequency. v = speed of sound in the air. v_D = detector speed. v_S = source speed (both relative to the air).
Assumptions	Sign rule: choose each sign so that motion **toward** the other raises f' ($+v_D$ on top, $-v_S$ on the bottom) and motion **away** lowers it. Speeds must be less than v. A reflection is two steps: wall as detector, then wall as source.
Use it for	Ambulances, trains, echoes off walls, whistles on a string ($v_S = r\omega$), and ranking observers.
Past questions	M25 Q16, Q24 · M24 Q7, Q14 · M22 Q16 · F25 Q5 · F23 Q11 · F22 Q16

Relativistic Doppler (light)

$$f' = f\sqrt{\frac{c \pm v}{c \mp v}}$$

Variables	v = relative speed between source and observer. Top signs = approaching (blueshift). Bottom signs = receding (redshift).
Assumptions	Light, with no medium. Only the relative speed matters. **This is on the midterm sheet and the F2025 final sheet but NOT on the posted final sheet**, so learn it.
Use it for	Galaxy redshift: convert with $f = \dfrac{c}{\lambda}$ to get $\left(\dfrac{\lambda'}{\lambda}\right)^2 = \dfrac{c+v}{c-v}$ for a receding source.
Past questions	M25 Q9 · F24 Q18 (conceptual)

Interference conditions

$$\text{Constructive: } \frac{\Delta L}{\lambda} = 0, 1, 2, \ldots \qquad \text{Destructive: } \frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \ldots$$

Variables	ΔL = path length difference.
Assumptions	The sources are **in phase** (no extra phase shifts). In between gives intermediate intensity, so compare which integer or half-integer is closer. Between two sources, ΔL can never be larger than their separation d.
Use it for	Is the point loud or quiet? Bright or dark? Is that path difference even possible?
Past questions	M22 Q7, Q9 · F23 Q16, Q17 · F22 Q8, Q10, Q11

String fixed at both ends / pipe open at both ends

$$f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}, \quad n = 1, 2, 3, \ldots$$

Variables	n = harmonic number, which equals the number of loops (half-wavelengths). L = string or pipe length. v = wave speed (string: $\sqrt{\frac{\tau}{\mu}}$. Pipe: speed of sound).
Assumptions	Both ends are the same type (nodes for a string, pressure nodes for an open pipe). **A pipe closed at both ends also uses this formula** (not stated on the sheet). All harmonics are present, so consecutive ones differ by f_1. Mode n has $n + 1$ nodes and n antinodes.
Use it for	Guitar strings, finding tension, flutes, identifying the pipe type from listed resonances, and the effect of changing L ($f \propto \frac{1}{L}$).
Past questions	M25 Q2, Q6, Q14 · M24 Q12, Q17–Q20 · M23 Q4, Q19 · M22 Q14 · F25 Q8 · F24 Q3 · F23 Q8, Q9, Q12 · F22 Q3

Pipe closed at one end

$$f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}, \quad n = 1, 3, 5, \ldots$$

Variables	Only **odd** n. The closed end is a displacement node (pressure antinode). The open end is a displacement antinode (pressure node).
Assumptions	Consecutive resonances differ by $2f_1$. Careful with wording: "f_3" means $n = 3$, but the "third mode" would be $n = 5$.
Use it for	Clarinets, organ pipes, and telling a closed–open pipe from an open–open one (the ratios to f_1 come out odd).
Past questions	M25 Q14, Q19 · M23 Q9 · F24 Q4

Optics: interference and diffraction of light

Double slit: bright fringes

$$d \sin\theta = m\lambda, \quad m = 0, 1, 2, \ldots$$

Variables	d = slit (or source) separation. θ = angle from the central axis. m = order ($m = 0$ is the central maximum).
Assumptions	Coherent sources in phase, with a far screen. For small angles the fringe position is $y_m = \dfrac{m\lambda D}{d}$ and the spacing is $\Delta y = \dfrac{\lambda D}{d}$ (**not on sheet**, D = screen distance).
Use it for	Fringe positions, two wavelengths landing at the same spot ($m_1\lambda_1 = m_2\lambda_2$), white-light colour order, and counting maxima around two sources ($\lvert\sin\theta\rvert \le 1$, so $\lvert m\rvert \le \frac{d}{\lambda}$, with 2 points per m on a full circle except at the ends).
Past questions	M25 Q18, Q25 · M23 Q15 · M22 Q10 · F24 Q6 · F23 Q10 · F22 Q12

Double slit: dark fringes

$$d \sin\theta = \left(m + \tfrac{1}{2}\right)\lambda, \quad m = 0, 1, 2, \ldots$$

Variables	Same as above.
Assumptions	Same as above.
Use it for	Positions of the dark fringes between bright ones.
Past questions	Used with the bright-fringe questions.

Thin film conditions

$$2t = m\frac{\lambda_0}{n}, \qquad 2t = \left(m - \tfrac{1}{2}\right)\frac{\lambda_0}{n}, \qquad m = 1, 2, 3, \ldots$$

Variables	t = film thickness (also written L). n = index of the **film**. λ_0 = wavelength in air. $2t$ = extra path of the ray that crosses the film twice.
Assumptions	Normal incidence. **Which equation is bright and which is dark depends on the reflection phase shifts** (table below). Net shift 0 (zero or two half-waves): $2t = m\dfrac{\lambda_0}{n}$ is bright and the other is dark. Net shift $\frac{1}{2}\lambda$: swap them. For **transmitted** rays (r_3, r_4), count only the reflections r_4 makes inside the film.
Use it for	Finding the visible wavelength that is bright or dark (try $m = 1, 2, 3$ and keep 400–700 nm), and the k-th smallest thickness.
Past questions	M23 Q16 · M22 Q11, Q12 · F24 Q5 · F22 Q13, Q14

Reflection phase shift table

$$\text{Off lower } n: 0, \qquad \text{off higher } n: \tfrac{1}{2}\lambda\ (=\pi)$$

Variables	"Off higher index" means the light is in a lower-n material and reflects from a higher-n one.
Assumptions	Applies to light (and is the same idea as a string going light to heavy). Transmitted light never gets a shift.
Use it for	The first step of every thin-film question: find the net shift.
Past questions	F25 Q14 · all thin-film questions

Single slit: dark fringes

$$a\sin\theta = m\lambda, \quad m = 1, 2, 3, \ldots$$

Variables	a = slit width. m starts at 1 (there is no dark fringe at the centre).
Assumptions	Far screen. For small angles, $y_m = \dfrac{m\lambda D}{a}$ (not on sheet). The central maximum is twice as wide as the side maxima.
Use it for	Slit width from fringe positions or a graph, distances between minima, and missing double-slit orders (set the two equations at the same θ: $\dfrac{d}{a} = m_{\text{missing}}$, not on sheet).
Past questions	M23 Q14, Q20 · M22 Q8 · F25 Q11, Q15 · F24 Q7, Q16 · F23 Q15 · F22 Q9

Double-slit intensity

$$I = 4I_0\cos^2\left(\tfrac{1}{2}\phi\right), \qquad \phi = \frac{2\pi d}{\lambda}\sin\theta$$

Variables	I_0 = intensity from one slit alone. ϕ = phase difference between the two waves at angle θ.
Assumptions	Narrow slits (single-slit diffraction ignored), equal amplitudes, coherent light.
Use it for	Intensity at a given angle, or comparing it to the maximum $4I_0$.
Past questions	Not directly tested. F24 Q30 is related (probability $\propto$ intensity).

Rayleigh criterion (resolution)

$$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$$

Variables	θ_R = smallest resolvable angle. d = diameter of the **circular** aperture (lens or mirror).
Assumptions	Circular aperture, diffraction-limited. A smaller λ or a larger d gives better resolution.
Use it for	Telescope or eye resolution, and why electron microscopes resolve better (smaller $\lambda = \dfrac{h}{p}$).
Past questions	F24 Q17, Q28

Modern physics (final exam only)

Coulomb force and potential energy

$$F_e = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}, \qquad U = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$$

Variables	e = elementary charge. r = electron–proton distance. ϵ_0 = permittivity of free space.
Assumptions	Point charges. For the hydrogen atom, U is actually **negative** ($-\frac{e^2}{4\pi\epsilon_0 r}$) because the charges are opposite. The sheet drops the sign.
Use it for	Bohr-model derivations (set $F_e = \frac{m_e v^2}{r}$).
Past questions	Not directly tested.

Potential energy from voltage

$$U = qV$$

Variables	q = charge (use e for an electron). V = potential difference.
Assumptions	An electron accelerated from rest through V gains $K = eV$.
Use it for	"What voltage gives the electron wavelength λ?": $p = \frac{h}{\lambda}$, then $K = \frac{p^2}{2m}$, then $V = \frac{K}{e}$.
Past questions	F25 Q25

Kinetic energy

$$K = \frac{1}{2}mv^2$$

Variables	Non-relativistic kinetic energy.
Assumptions	$v \ll c$. Combine with $p = mv$ to get $K = \frac{p^2}{2m}$ (**not on sheet**), which you need constantly.
Use it for	de Broglie problems, and comparing electrons with neutrons.
Past questions	F25 Q16, Q25 · F24 Q19, Q29 · F22 Q29

Photon energy

$$E = hf$$

Variables	h = Planck's constant $= 6.63 \times 10^{-34}$ J·s $= 4.136 \times 10^{-15}$ eV·s. f = frequency.
Assumptions	Light comes in quanta. Use $E = \frac{hc}{\lambda}$, with $hc = 1240$ eV·nm (not on sheet).
Use it for	Photoelectric effect, emission lines, photon counting, and Compton energies.
Past questions	F25 Q17, Q20 · F24 Q21, Q25, Q26 · F23 Q19, Q21, Q27 · F22 Q21, Q24, Q28

de Broglie relation / photon momentum

$$p = \frac{h}{\lambda}$$

Variables	p = momentum. For a particle $p = mv$. For a photon $p = \frac{E}{c}$ (not on current sheet).
Assumptions	Holds for both photons and matter waves.
Use it for	Electron wavelengths, matching an electron to an X-ray, recoil of an atom absorbing a photon (momentum conservation), and resolution of electron microscopes.
Past questions	F25 Q16, Q19, Q25 · F24 Q19, Q26, Q28, Q29 · F22 Q24, Q26, Q29

Photoelectric effect

$$hf = K_{max} + \Phi, \qquad K_{max} = eV_{stop}$$

Variables	Φ = work function, the minimum energy to free an electron (eV). K_{max} = kinetic energy of the fastest electron. V_{stop} = stopping voltage.
Assumptions	One photon is absorbed by one electron, instantly (no lag). If $hf < \Phi$, **no current at any intensity**. Intensity changes the current, not K_{max}.
Use it for	Identifying a metal, threshold questions, comparing lasers, and conceptual true/false questions.
Past questions	F25 Q20 · F24 Q25 · F23 Q21 · F22 Q17, Q23

Centripetal acceleration

$$a_c = \frac{v^2}{r}$$

Variables	v = orbital speed. r = radius.
Assumptions	Uniform circular motion.
Use it for	Bohr-model orbits.
Past questions	Not directly tested.

Compton shift

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$$

Variables	m = mass of the **electron** ($\frac{h}{m_e c} = 2.43$ pm, the Compton wavelength). ϕ = photon scattering angle. $\Delta\lambda = \lambda' - \lambda$.
Assumptions	Photon scattering off a free electron at rest. The shift does not depend on the incoming λ. It is 0 at 0° and largest ($\frac{2h}{mc}$) at 180°. The electron gains $K = E - E'$.
Use it for	Finding λ', ranking angles, finding the angle from the electron's energy, and energy-loss percentage questions.
Past questions	F25 Q23, Q24 · F24 Q27 · F23 Q20, Q29 · F22 Q19, Q20, Q28

Rydberg formula

$$\frac{1}{\lambda} = R\left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2}\right)$$

Variables	$R = 1.09737 \times 10^7$ m^{-1} (Rydberg constant, not the gas constant!). n = energy level numbers.
Assumptions	Hydrogen only (one electron, $Z = 1$).
Use it for	Wavelength of a hydrogen spectral line. The smallest jump gives the longest wavelength.
Past questions	F25 Q17 · F24 Q24

Photon emission rate

$$R = \frac{P}{hf}$$

Variables	R = photons per second (again, not Rydberg or the gas constant). P = power.
Assumptions	Monochromatic light.
Use it for	"How many photons per second hit the panel?" (with $P = IA$).
Past questions	F23 Q19

Heisenberg uncertainty

$$\Delta x \, \Delta p_x \geq \frac{h}{2\pi}$$

Variables	Δx = position uncertainty. Δp_x = momentum uncertainty.
Assumptions	**Check your sheet:** the posted sheet and the F2022–F2024 sheets use $\frac{h}{2\pi}$, while F2025 used $\frac{h}{4\pi}$. Use whichever is printed. $\Delta p = m\Delta v$ (not on sheet).
Use it for	Minimum position or speed uncertainty, comparing particles of different mass, and ratio questions (doubling Δx halves Δp).
Past questions	F25 Q19 · F24 Q20 · F23 Q28 · F22 Q22, Q27

Bohr model of hydrogen

$$a_0 = \frac{\epsilon_0 h^2}{\pi m_e e^2}, \qquad E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2} = \frac{-13.6 \text{ eV}}{n^2}$$

Variables	a_0 = Bohr radius (0.529 Å). E_n = energy of level n (negative means bound). $n = 1$ is the ground state, so the "k-th excited state" is $n = k + 1$.
Assumptions	Hydrogen (one electron). For a hydrogen-like system, use the constant given in the question.
Use it for	Absorption and emission between levels ($\Delta E = E_{high} - E_{low}$), which jump gives the longest wavelength, and finding n_{low}.
Past questions	F25 Q17 · F24 Q24 · F23 Q27

Infinite square well: energy

$$E_n = \frac{h^2}{8 m_e L^2} n^2$$

Variables	L = well width. $n = 1, 2, 3, \ldots$ m_e = particle mass (an electron unless told otherwise).
Assumptions	Infinitely high walls, so $\psi = 0$ at the walls. $E \propto n^2$ and $E \propto \frac{1}{L^2}$.
Use it for	Width from an energy, emission wavelengths ($\Delta E = (n_i^2 - n_f^2)E_1$), ranking energy gaps, and the effect of changing L.
Past questions	F25 Q18 · F24 Q21 · F23 Q23, Q24, Q26, Q30

Infinite square well: wave function

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)$$

Variables	$\sqrt{\frac{2}{L}}$ = normalization constant. Inside the well, $0 \le x \le L$.
Assumptions	Already normalized. State n has n humps (most likely positions at $\frac{L}{2n}, \frac{3L}{2n}, \ldots$) and $n+1$ nodes counting the walls.
Use it for	Choosing the right ψ for a state, the most likely positions, and counting nodes to find n.
Past questions	F25 Q21 · F24 Q23 · F23 Q23

Probability

$$P(x_1, x_2) = \int_{x_1}^{x_2} |\psi(x)|^2 dx$$

Variables	$\lvert\psi\rvert^2$ = probability density.
Assumptions	ψ is normalized: $\int \lvert\psi\rvert^2 dx = 1$ over all space (not on sheet). For a very narrow strip, $P \approx \lvert\psi(x)\rvert^2\, dx$.
Use it for	Probability in part of the well, using $\int \sin^2(ax)\, dx = \frac{x}{2} - \frac{\sin(2ax)}{4a}$ from the math section, and normalization constants.
Past questions	F25 Q22 · F24 Q22 · F23 Q25 · F22 Q30

Hydrogen ground state

$$\psi(r) = \frac{1}{\sqrt{\pi}\, a_0^{\frac{3}{2}}} e^{-\frac{r}{a_0}}, \qquad P(r) = \frac{4}{a_0^3} r^2 e^{-\frac{2r}{a_0}}$$

Variables	$\psi(r)$ = 1s wave function. $P(r)$ = **radial** probability density (probability per unit radius). a_0 = Bohr radius.
Assumptions	Ground state only ($n = 1$). The most probable radius is $r = a_0$.
Use it for	Probability of finding the electron between two radii: $\int P(r)\, dr$. Also where it is most likely.
Past questions	Not tested in past exams.

Tunneling

$$T \approx e^{-2bL}, \qquad b = \sqrt{\frac{8\pi^2 m_e (U_b - E)}{h^2}}$$

Variables	T = transmission probability. L = barrier thickness. U_b = barrier height. E = particle energy.
Assumptions	Thick or high barrier ($T \ll 1$) and $E < U_b$. Compare barriers using $bL \propto \sqrt{U_b - E} \times L$: a smaller bL gives a larger T.
Use it for	Ranking barriers, and the effect of thickness or height.
Past questions	F23 Q22

Schrödinger equation

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0, \quad k = \frac{2\pi\sqrt{2m_e(E-U)}}{h}; \qquad \frac{d^2\psi}{dx^2} + \frac{8\pi^2 m_e}{h^2}\big(E - U(x)\big)\psi = 0$$

Variables	k = wavenumber of the matter wave. $E - U$ = kinetic energy.
Assumptions	Time-independent, one dimension. Where U goes up, k goes down, so $\lambda = \dfrac{2\pi}{k}$ goes up.
Use it for	Conceptual questions about wavelength or wavenumber in regions of different potential.
Past questions	F24 Q19

Constants and math formulas: when they show up

Gas and sound constants

$$R = 8.314, \quad M_{air} = 28.8\ \tfrac{\text{g}}{\text{mol}}, \quad \gamma_{mono} = \tfrac{5}{3}, \quad \gamma_{di} = \tfrac{7}{5}, \quad T_K = T_{^\circ\text{C}} + 273.15$$

Variables	Use M in kg·mol^{-1} (0.0288) in $v = \sqrt{\dfrac{\gamma RT}{M}}$. Air is diatomic.
Use it for	Speed of sound in gases.
Past questions	M25 Q8 · F24 Q14

Sound and light constants

$$v_{sound} = 343\ \tfrac{\text{m}}{\text{s}}, \quad I_0 = 10^{-12}\ \tfrac{\text{W}}{\text{m}^2}, \quad c = 3.00 \times 10^8\ \tfrac{\text{m}}{\text{s}}, \quad g = 9.81\ \tfrac{\text{m}}{\text{s}^2}$$

Variables	343 is for air at 20 °C. Questions often give a different value: use theirs.
Use it for	Doppler, pipes, dB, and pendulums.
Past questions	Throughout

Modern physics constants

$$h,\ e,\ m_e,\ m_p,\ a_0,\ \epsilon_0,\ k_B,\ \sigma,\ R_{Rydberg},\ 1\,u$$

Variables	e converts J to eV (1 eV = 1.602×10^{-19} J). σ is the Stefan–Boltzmann constant. **Stefan's law $P = \sigma A T^4$ is not on the current sheet** (it was on F2022).
Use it for	Every modern physics calculation.
Past questions	F22 Q25 (Stefan)

Sum-to-product identities

$$\cos\alpha \pm \cos\beta = 2\cos\left(\tfrac{1}{2}(\alpha \pm \beta)\right)\cos\left(\tfrac{1}{2}(\alpha \mp \beta)\right), \qquad \sin\alpha \pm \sin\beta = 2\sin\left(\tfrac{\alpha\pm\beta}{2}\right)\cos\left(\tfrac{\alpha\mp\beta}{2}\right)$$

Assumptions	Equal amplitudes only.
Use it for	Adding two equal waves or oscillations by hand (beats, superposition, standing waves).
Past questions	M24 Q8 · F24 Q12

Phasors (complex amplitude)

$$\tilde{A} = Ae^{j\theta} = A\big(\cos\theta + j\sin\theta\big)$$

Variables	A = amplitude (length). θ = phase (angle). Real part = x component. Imaginary part = y component.
Assumptions	Same frequency for every wave being added. When finding the angle with $\tan^{-1}$, **add 180° if the real part is negative**.
Use it for	Adding waves with **different** amplitudes or several phases, finding a phase from Cartesian components, and the complex amplitude of an SHM.
Past questions	M25 Q7, Q15 · M23 Q1 · F25 Q7 · F24 Q8 · F23 Q5

Angle identities

$$\cos(\alpha \pm \beta),\ \ \sin(\alpha \pm \beta),\ \ \sin(-\alpha) = -\sin\alpha,\ \ \cos\left(\alpha \pm \tfrac{\pi}{2}\right) = \mp\sin\alpha,\ \ \sin\left(\alpha \pm \tfrac{\pi}{2}\right) = \pm\cos\alpha$$

Use it for	Rewriting sin as cos (and back) to match answer choices, and turning a minus sign into a phase of π.
Past questions	M22 Q3 · F24 Q10 · F22 Q6

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Assumptions	Keep the root that is physical or listed as an option.
Use it for	Solving $T_e = \dfrac{4r}{(1+r)^2}$ for an impedance ratio.
Past questions	F24 Q2

Integrals and power reduction

$$\int \sin^2(ax)\,dx = \frac{x}{2} - \frac{\sin(2ax)}{4a} + c, \qquad \sin^2\theta = \tfrac{1}{2}\big(1 - \cos 2\theta\big)$$

Use it for	Probability in an infinite well, where $a = \dfrac{n\pi}{L}$.
Past questions	F25 Q22 · F24 Q22